

Section-A (15)

Attempt all the questions from this section)

Question-1: Choose the correct option:

1×15=15

- i) The compound interest on ₹8000 at 5% per annum for 2 years is
 - (a) ₹800
 - (b) ₹820
 - (c) ₹840
 - (d) ₹800.50
- ii) The perimeter of a rectangle is 60 cm and its length is 20 cm, then its breadth is
 - (a) 10 cm
 - (b) 15 cm
 - (c) 20 cm
 - (d) 25 cm
- iii) The mean of first 10 natural numbers is
 - (a) 5
 - (b) 5.5
 - (c) 6
 - (d) 4.5
- iv) The graph of a frequency polygon is formed by joining
 - (a) class limits
 - (b) class marks
 - (c) frequencies
 - (d) cumulative frequencies
- v) Area of a triangle with base 10 cm and height 6 cm is
 - (a) 30 cm²
 - (b) 60 cm²
 - (c) 15 cm²
 - (d) 20 cm²
- vi) The circumference of a circle of radius 7 cm is
 - (a) 14π cm
 - (b) 49π cm
 - (c) 21π cm
 - (d) 7π cm
- vii) The volume of a cube of side 3 cm is
 - (a) 9 cm³
 - (b) 18 cm³
 - (c) 27 cm³
 - (d) 81 cm³
- viii) $\sin 30^\circ$ is equal to
 - (a) 1
 - (b) $\sqrt{3}/2$
 - (c) $1/2$
 - (d) $\sqrt{2}/2$
- ix) The distance between points (0,0) and (3,4) is
 - (a) 5 units
 - (b) 7 units
 - (c) 4 units
 - (d) $\sqrt{7}$ units
- x) The class mark of the class interval 20–30 is
 - (a) 20
 - (b) 25
 - (c) 30
 - (d) 10

- xi)** The point $(0, -5)$ lies on
 (a) x-axis
 (b) y-axis
 (c) origin
 (d) neither axis
- xii)** The area of a triangle with vertices $(0,0)$, $(4,0)$, $(0,3)$ is
 (a) 6 sq units
 (b) 12 sq units
 (c) 24 sq units
 (d) 7 sq units
- xiii)** A cuboid has volume 120 cm^3 and base area 24 cm^2 . Its height is
 (a) 3 cm
 (b) 4 cm
 (c) 5 cm
 (d) 6 cm
- xiv)** The total surface area of a cuboid of dimensions l, b, h is
 (a) $lb + bh + hl$
 (b) $2(lb + bh + hl)$
 (c) lbh
 (d) $2lbh$
- xv)** The solution of $x = 2, y = -1$ lies in which quadrant?
 (a) First
 (b) Second
 (c) Third
 (d) Fourth

Question-2).

[13]

- i) Find the amount and the compound interest on Rs 24000 for 18 months at 10% per annum, the interest being compounded half-yearly.
- ii) Calculate the side of a rhombus if its diagonals are 18 cm and 24 cm respectively.
- iii) Draw a quadrilateral ABCD with $AB=6 \text{ cm}$, $BC=4 \text{ cm}$, $CD=4 \text{ cm}$, and $\angle ABC = \angle BCD = 90^\circ$.

Question-3).

[12]

- i) The base of an isosceles triangle measures 80 cm and its area is 360 cm^2 . Find the perimeter of the triangle.
- ii) A wire when bent in the form of an equilateral triangle encloses an area of $121\sqrt{3} \text{ cm}^2$. If the same wire is bent in the form of a circle, find the area of the circle.
- iii) The volume of a rectangular solid is 3600 cm^3 . If it is 20 cm long and 9 cm high, find its width.

SECTION-B

($10 \times 4 = 40$)

(Attempt any four questions from this section)

Question-4)

- i) Construct a frequency polygon for the following data:

Class	0-10	10-20	20-30	30-40	40-50
Frequency	3	7	6	2	5

- ii) Madhu practiced on her sitar 45 minutes, 30 minutes, 60 minutes, 50 minutes and 20 minutes. Out of a total of 16 observations, ascending order, the 8th and 9th observations are 25 and 27. What is her mean practice time? What is the median?
- iii) Out of a total of 16 observations, ascending order, the 8th and 9th observations are 25 and 27. What is the median?

Question-5)

- i) A room is 18 m long 14 m wide in 8 m high. Find the area of the four walls and the total surface area of the six faces of the room.
- ii) The area of the trapezium is 240 m^2 and the sum of the parallel sides is 48 m, find the height.
- iii) Find the area of the triangle whose sides are 26 cm, 28 cm and 30 cm.

Question-6)

- i) If x is an acute angle and $\sin x = \cos x$, find the value of $2\tan 2x + \sin 2x - 1$.
- ii) Evaluate $\sec 50^\circ \cdot \sin 40^\circ + \cos 40^\circ \cdot \operatorname{cosec} 50^\circ$.

Question-7)

- i) Solve graphically $x+y=5$, $x-y=1$.
- ii) If $\tan x = a$, find the value of $\cos x + \sin x / \cos x - \sin x$.
- iii) The distance between two points $(0,0)$ and $(x,3)$ is 5, find x .

Question-8)

- i) Two opposite angles of a parallelogram are $(3x-2)^\circ$ and $(50-x)^\circ$. Find the measure of each angle of the parallelogram.
- ii) Find the length of the side of the triangle whose vertices are $A(3,4)$, $B(8,-1)$ and $C(4,-6)$.
- iii) The present price of a scooter is Rs. 7290. If its value decreases every year by 10%, then what was its value 3 years ago?
- iii(b) A sum of money becomes ₹7,744 in 2 years at compound interest, at the rate of 4% per annum. Find the principal.